

Exercise 3: Calculated Fields and Aggregation

Three worksheets, three captions. No memo.

Item	Details
Points	100
Data	clean/mvu_student_dim.csv, clean/mvu_enrollment_fact.csv, clean/mvu_term_gpa.csv
Reading	Learning Tableau 2025 (Milligan, 6th ed.), free through UTD Libraries: Ch 4 and Ch 5
Walkthrough	Module 3 of the MVU Tableau Walkthrough follows these same steps, with checkpoints.
Submit	One packaged workbook (.twbx) with Worksheets 1–3, a visible caption on each, and every calculated field named, formatted and commented. No memo.

Worksheet 1: row-level versus aggregate

1. Connect mvu_student_dim.csv. Create Aid per Credit (row) = [total_aid_awarded] / [credits_attempted].
2. Create Aid per Credit (agg) = SUM([total_aid_awarded]) / SUM([credits_attempted]).
3. college on Rows; both fields on Columns, the first set to Average. Format both as currency with no decimals. Name it Worksheet 1.

Checkpoint: Aid per Credit (agg) across all students, to the nearest dollar?

Worksheet 2: academic standing

1. Connect mvu_term_gpa.csv. Create Academic Standing with IF / ELSEIF, thresholds from high to low: 3.70 Excellent, 3.00 Good standing, 2.00 Academic warning, else Probation.
2. Academic Standing on Rows; student_id as Count (Distinct) on Columns. Name it Worksheet 2.
3. Add up the bars and compare with the number of students.

Checkpoint: Distinct students who were ever on Probation?

Worksheet 3: the summary cross-tab

Build these fields first. Relate the three files on student_id.

1. Last Term Attended = { FIXED [student_id] : MAX([term_start_date]) }
2. Months to Completion = IF [status] = "Graduated" THEN DATEDIFF('month', [entry_date], [Last Term Attended]) END
3. Graduation Rate = AVG([is_graduated]), with a default format of Percentage, 1 decimal.

Checkpoint: Overall graduation rate across all 1,400 students (percentage, one decimal)?

4. Build the cross-tab: college and program_name on Rows; distinct students, average term_gpa, Graduation Rate and median Months to Completion. Sort by Graduation Rate. Name it Worksheet 3.

5. Duplicate it, filter cohort_year to 2020 and 2021, and re-sort. Name it Worksheet 3 — 2020–21 Cohorts.

Checkpoint: Public Health's graduation rate for the 2020 and 2021 cohorts only (percentage, one decimal)?

Your three captions

Each worksheet carries a caption: a few sentences answering its question, written while the view is in front of you. Turn it on with Worksheet > Show Caption, then double-click below the view to edit. Where a worksheet has two versions, put the caption on the second one.

Sheet	What your caption must answer
Worksheet 1	Why are the two aid-per-credit numbers different? Which answers “what does a typical student receive per credit,” and which answers “what does MVU spend per credit”?
Worksheet 2	Why do the bars add up to more students than exist? Which cutoff would you defend changing, and what would that do to the chart?
Worksheet 3	Which programs stand out on graduation rate, and how much of that ranking is explained by cohort composition rather than the program itself? What changed when you restricted to 2020–21?

Worksheet 3's caption carries 25% on its own. Naming the top and bottom programs from the unfiltered table does not earn full marks. Noticing that the ranking was mostly cohort mix is the point of the exercise.

Before you submit: open each sheet and confirm the caption is showing. A caption toggled off still holds your text but won't be visible when your instructor opens the workbook.

Submit for Exercise 3

- Worksheet 1: row-level versus aggregate aid per credit, by college
- Worksheet 2: distinct students by Academic Standing
- Worksheet 3: the summary cross-tab with at least three calculated fields, plus the 2020–21 version
- A visible caption on each
- Packaged workbook (.twbx) with every calculated field named, formatted and commented. No memo.